

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Electrical Engineering
2. Subject Name: Measurements and Electronic Instruments L-T-P: 3-0-0
Credits:3
3. Pre-requisites: None
4. Syllabus and reference books:

Syllabus:

Instrument specifications and error analysis. Average, RMS value measurement; rectifier-based measurements; range extension of meters. Measurement of low and high resistances. Sensitivity of dc bridge, principles of ac bridges for measurement of L and C. Instrument transformers: CT, PT. Hall effect sensor, clamp-on meter. Potentiometers. Capacitive and inductive interference, shielding.

Essential elements of electronic instruments. Analog instruments: Analog DC voltmeters, ohmmeter, analog AC voltmeters, precision rectifier voltmeter, electronic multimeter and different probes.

Basics of digital measurements: scale of-2000 Counter, BCD-to-seven segment decoder, digital displays, analog-to-digital conversion, sample and hold, linear-ramp ADC, digital-ramp ADC, digital-to-analog converter. Digital instruments: single-slope ramp generator type digital voltmeter, dual slope integrator based digital voltmeter, digital ohmmeter, digital AC voltmeter and digital multimeter, digital frequency meter, accuracy measurements, reciprocal counting based frequency meter, pulse-width measurement, wave-shaping circuit.

Signal and function generators: basic theory for oscillation, Low frequency signal generator, Wein Bridge Oscillators, triangular wave generators, square wave generator, sine wave generator.

Digital Storage Oscilloscope: Operation, Storing a waveform, Retrieval from the waveform, Flat screen display, Memory and resolution, Interpolation, Sampling rate and bandwidth.

Reference Books:

- 1) Electronic Instrumentation and Measurements, David A. Bell, 3rd Edition, Oxford University Press India
- 2) Electrical Measurements & Instrumentation, U.A. Bakshi, A. V. Bakshi, Technical Publications
- 3) Electronic Instrumentation, H. S. Kalsi, 3rd Edition, McGraw Hill Education
- 4) Linear Integrated Circuits by D. Roy Choudhury and Shail B. Jain, New Age International Publisher

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1	Instrument specifications and error analysis, accuracy, precision, sensitivity, resolution.	2
2	Average, RMS value measurement; rectifier-based measurements; range extension of meters.	1
3	Measurement of low and high resistances.	2
4	Sensitivity of dc bridge, Principles of ac bridges for measurement of L and C.	4
5	Instrument transformers: CT, PT.	3
6	Hall effect sensor, clamp-on meter.	1
7	Potentiometers.	1
8	Capacitive and inductive interference, shielding.	3
9	Essential elements of electronic instruments: comparator, op-amp, difference amplifier, development of instrumentation amplifier.	1
10	Analog instruments: Analog DC voltmeters and different configurations, op-amp based voltmeter, series ohmmeter, shunt ohmmeter, linear ohmmeter, analog AC voltmeters, precision rectifier voltmeter, electronic multimeter and different probes.	4
11	Basics of digital measurements: scale of-2000 Counter, BCD-to-seven segment decoder, digital displays, analog-to-digital conversion, sample and hold, linear-ramp ADC, digital-ramp ADC, digital-to-analog converter.	3
12	Digital instruments: single-slope ramp generator type digital voltmeter, dual slope integrator based digital voltmeter, digital ohmmeter, digital AC voltmeter and digital multimeter, digital frequency meter, accuracy measurements, reciprocal counting based frequency meter, pulse-width measurement, wave-shaping circuit	4
13	Signal and function generators: basic theory for oscillation, Low frequency signal generator, Wein Bridge Oscillators, triangular wave generators, square wave generator, sine wave generator.	3
14	Digital Storage Oscilloscope: Operation, Storing a waveform, Retrieval from the waveform, Flat screen display, Memory and resolution, Interpolation, Sampling rate and bandwidth.	4
Total number of hours		36